

Advanced Machine Learning Model Comparison on Titanic Survival Prediction

Problem Statement

The objective of this project was to predict whether a passenger survived the Titanic disaster using machine learning classification algorithms. The project aimed to compare multiple advanced machine learning models and identify the best-performing model based on evaluation metrics.

Dataset Used

The Titanic Survival Dataset was used for this project. The dataset contains passenger information such as:

- Passenger Class (Pclass)
- Sex
- Age
- Fare
- Survival Status (Target Variable)

The dataset was cleaned by handling missing values and encoding categorical variables before model training.

Algorithms Used

The following machine learning algorithms were implemented:

1. Logistic Regression
2. Decision Tree Classifier
3. Random Forest Classifier
4. Gradient Boosting Classifier

Additionally, GridSearchCV was used to perform hyperparameter tuning on the Random Forest model.

Model Performance

Logistic Regression

- Accuracy: 79.89%

Decision Tree

- Accuracy: 74.86%

Random Forest

- Accuracy: 80.45%

Gradient Boosting

- Accuracy: 81.01%

Tuned Random Forest

- Best Parameters:
 - max_depth = 10
 - n_estimators = 200
- Cross Validation Score: 83.29%

Challenges Faced

- Handling missing values in the Age column.
- Converting categorical variables such as Sex into numerical values.
- Understanding the differences between multiple classification algorithms.
- Selecting appropriate hyperparameters for model optimization.

Key Findings

- Decision Tree produced the lowest accuracy and was more prone to overfitting.
- Random Forest improved prediction performance by combining multiple decision trees.
- Gradient Boosting achieved the highest testing accuracy of 81.01%.
- Hyperparameter tuning further improved Random Forest performance.
- Ensemble learning methods generally performed better than a single Decision Tree model.

Conclusion

The project demonstrated the effectiveness of advanced machine learning techniques for classification tasks. Gradient Boosting emerged as the best-performing model on the test dataset, while the tuned Random Forest achieved the strongest cross-validation performance.

This highlights the importance of model selection and hyperparameter optimization in building accurate predictive systems.